Optimizing an investment portfolio under Solvency II

It does indeed solve an important problem

Maastricht University
Bachelor thesis in Econometrics & Operations Research
Name: Florian Koenigstorfer
Student ID: i6080626
Supervisor: Professor Dr. Antoon Pelsser
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Abstract

The goal of this thesis is to maximize the expected surplus of the investment portfolio of a life insurance company under the restrictions imposed by Solvency II. The problem is modeled using a linear programming approach, where the restrictions from Solvency II are phrased as constraints.

The first section of the analysis reveals that the optimal allocation is unstable when there is little information about the development of the economy in different scenarios. However, as the amount of information increases, the optimal allocation becomes stable. Further analysis reveals that when the mortality increases, it threatens to bankrupt the insurer. An increase to the assets of the insurer protects the company from bankruptcy and ensures its survival. Analyzing the optimal allocation for an investment universe including stocks as well as bonds reveals that the proportion of each asset class in the portfolio changes, depending on the size of the investment budget. For a low budget, the proportion of bonds is high and the proportion of stocks in the portfolio is low. As the investment budget increases, the proportion of stocks tends to increase, while the proportion of bonds tends to decrease.

1 Introduction

In November 2009, the European Union adopted the Solvency II framework [2], a new set of regulation attempting to harmonize insurance regulation within the EU. Within this set of regulations, the EU imposes a set of restrictions on the investments an insurance company can take, in order to reduce the risk of insolvency. However, these restrictions pose certain challenges for insurance companies when they are designing an investment strategy. In a market analysis, the advisory company WillisTowersWatson found that "Solvency II valuations are generally complex, time consuming and potentially expensive." [7] The calculations require "the involvement of multiple departments and teams, and a plethora of individual and manual tasks. Together, these can lead to very inefficient processes, with expensive actuaries often carrying out routine but laborious tasks." [7] This thesis attempts to simplify this process by investigating the optimal investment strategy under the restrictions imposed by the new Solvency II framework. An investment is considered optimal, if it not only satisfies the constraints imposed by Solvency II, but also maximizes the expected surplus one time unit ahead.
The first section provides a detailed description of the problem from an economic perspective. Not only will this section detail the components of the insurer's balance sheet, it will also explain the impact of Solvency II on the estimation of the Expected Surplus. this section will also include the constraints the investor's balance sheet will be exposed to.
The following section of this paper concerns itself with the reformulation of the problem detailed in the section beforehand in a linear programming context. The section details the construction of the objective function as well as the constraints. The section also discusses how the uncertainty of the returns associated with the development of the value of the assets is addressed.
The fourth section of this paper focuses on the implementation of a very simplified model and discusses the results. The section not only aims to illustrate the setup of the model, but serves as a basis for future results to build upon.
On the basis of this simplified model, the fifth section explores the expansion from the simple model to a more realistic model. First of all, the section will include steps to make the optimal allocation more stable. Second of all, a shock to the liability side will be implemented. Finally, the section investigates what will happen if a second asset class is introduced to the investment universe.
The sixth section suggests how the problem could be improved.
The final section concludes the analysis and provides food for further thought.

2 The problem in more detail

This section outlines the economic intuition behind the setup of the model.

2.1 The Balance Sheet

The starting point of the model is a balance sheet of an insurance company. On the one hand, an insurer owns a portfolio of assets, which he selects from an investment universe Y. This investment universe can consist of not only stock and bonds, but also of real estate and currency. In this paper, the asset side of the Balance Sheet is represented by the product $c^{\mathrm{T}}Y$, where c represents the number of units of each asset in the investment universe of the insurer's portfolio.
On the other hand, the insurer also has liabilities to his clients, denoted by Z.
The difference between the assets and the liabilities yields the surplus. The surplus is the amount of money the insurance company has left, after it has paid its liabilities in full. These funds can be used to absorb shocks resulting from an unanticipated negative economic environment.
The value of the portfolio as well as the liabilities is its value at one of two distinct points in time: The present ($T = 0$) or one time unit ahead ($T = 1$). Table 1 below shows what the Balance Sheet at time $T = 0$ would look like.

Market Value of Assets $c^{\mathrm{T}}Y_{\mathrm{T=0}}$	Surplus $S_{\mathrm{T=0}} = c^{\mathrm{T}}Y_{\mathrm{T=0}} - Z_{\mathrm{T=0}}$
	Market Value of Liabilities $Z_{\mathrm{T=0}}$

Table 1: Balance Sheet at $T = 0$

2.1.1 Liabilities

People want to be covered for various types of eventualities. These can represent risks to their homes, their heath or even their lives. The desire to insure these risks represents a business opportunity for insurance companies. Due to the difference in the size and the distribution of the underlying claim(s), the liability side of each one of the insurances has to be modeled differently. For simplicity, this thesis focuses on the liabilities of a life insurer. Naturally, the same analysis can be replicated for the liabilities of any other kind of insurance.

2.1.2 Investments

Apart from insurance business itself, an insurance company has the opportunity to invest the premia from its clients until the underlying claim has to be paid out. An insurance company faces many different investment possibilities. While they can consider publicly traded assets such as stocks, corporate- and government bonds or currency, insurance

companies also have the opportunity to invest in private equity and beyond. In a White Paper, the OECD found that "in most OECD countries that provided information for 2008, bonds [...] remain by far the dominant asset class across life, non-life and composite insurance segments." [5] For the purpose of this paper, the investment universe will consist bonds and stocks.

2.2 Implications of Solvency II

2.2.1 Risk constraints via pre-defined shocks to elements of the balance sheet

During the turmoil of the recent financial crisis, the insurance sector was impacted strongly. Dr. Sebastian Schich, principal economist for the OECD, found that "the financial crisis has [...] had an increasingly visible impact on the insurance industry." [6] The OECD found that "[a] main channel through which insurance undertakings were affected by the market turmoil was via their asset side investments in equity and debt instruments as well as structured finance products." [5] In order to ensure the solvency of the insurance industry, even in the event of an economic downturn, the European Union implemented the Solvency II framework.
One of the main components of the Solvency II framework is the implementation of a so called standard formula [2]. This formula is used to calculate the solvency capital requirement for the investments of the insurer. An important element of the standard formula are a set of pre-defined shocks to certain risk components of a life insurer's business. Those risk components can affect one, or both, the asset side as well as the liability side of the balance sheet. The components of this formula are shown in Figure 1 [3] below.

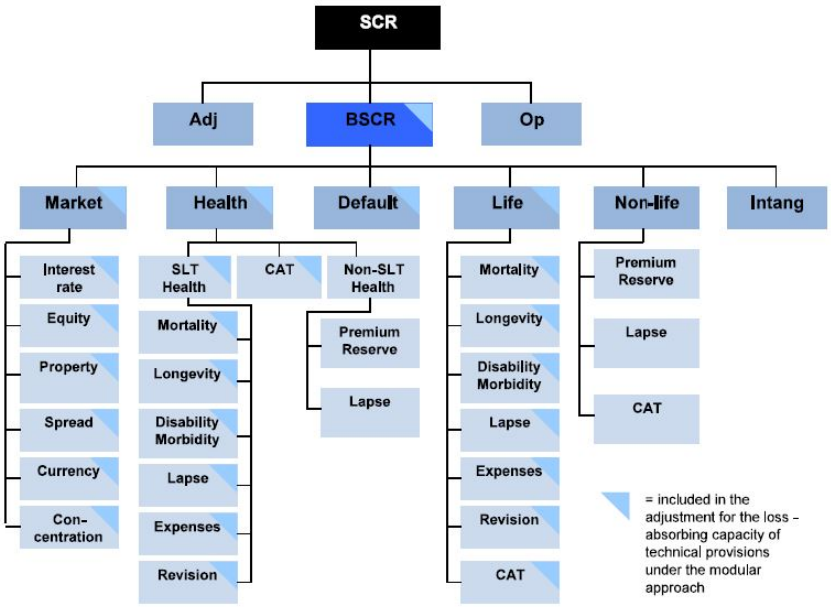

Figure 1: Overview over Risk Categories

In the context of the thesis, the shocks will be used to form restrictions on the investments the insurer can make. Since the main focus of the thesis lies on the optimization of an investment portfolio, the Market Risk component in Figure 1 will be of crucial importance. More specifically, since this thesis only considers a portfolio consisting of bonds and equity, the interest rate shock as well as the shock to equity will be the primary drivers on effecting the asset side.

Even though the focus of this thesis is the balance sheet of a life insurance company, the shocks to the liability side should be considered. Since the bulk of the thesis focuses on the optimization of the asset side, only a single shock will be implemented, namely the mortality shock. Since different types of assets have different types of risks and volatilities attached to them, Solvency II defines distinct shocks for each one of them. The advisory company Deloitte [4] summarizes these shocks in Figure 2 below.

Figure 1: Overview of Stress Factors Standard Model

INVESTMENT CLASS	SUB-RISK-MODULE	STRESS IN %	ADJUSTMENT
Shares Type I	Equity	39%	range +/- 10%
Shares Type II	Equity	49%	range +/- 10%
Private Equity	Equity	49%	range +/- 10%
Hedge Funds	Equity	49%	range +/- 10%
Infrastructure Equity Capital	Equity	30%	n. a.
Strategic Participations	Equity	22%	-
Property	Real Estate	25%	-
EU-Member State Bonds	Spread	0 %	-
Corporate Bonds	Spread	Dependency on Rating and Maturity	-
Covered Bonds	Spread	Relief vs. Corporate Bonds	-
Infrastructure Foreign Capital (leverage)	Spread	Relief vs. Corporate Bonds	-

Figure 2: Stress Factors

The graph doesn't reflect the correlation between different shocks. Figure 3 [4] below shows the correlation between different stresses.

Figure 2: Correlation Matrix Market Risk Module

CorrMkt	Interest Rate	Shares	Property	Spread	Exchange Rate	Concentration
Interest Rate	1					
Shares	A*	1				
Property	A*	0.75	1			
Spread	A*	0.75	0.5	1		
Exchange rate	0.25	0.25	0.25	0.25	1	
Concentration	0	0	0	0	0	1

A* the correlation amounts to 0 as far as the interest rate rise scenario in the interest rate risk module represents the adverse scenario, otherwise 0.5

Figure 3: Correlation Matrix

Solvency II specifies that an insurance company needs to remain solvent following each

of these shocks. This implies that after each one of the shocks, the assets of the insurer need to remain higher than its liabilities.

3 Rephrasing the problem in a LP context

3.1 Surplus at $T = 1$ and Objective Function

At time $T = 0$, we assume that the "economic state" is known. This means that, for instance, we can observe the interest rate term structure as well as the market prices of all assets and what value we need to use for the rate of mortality. The development of the economy during the time step from $T = 0$ to $T = 1$, however, is very much uncertain. For instance, the expected interest term structure my change at any point within a year. In order to incorporate this uncertainty in the model, the expected surplus at time $T = 1$ does not only take into account the economic situation at $T = 0$, but also a number of alternative economic scenarios. The expected surplus is therefore defined as the difference between the average value of the insurer's portfolio, at $T = 1$, and the average value of the insurer's liabilities, also at $T = 1$. Given that there are N distinct economic scenarios, the Balance Sheet at $T = 1$ can be expressed as in Table 2 below.

$\frac{1}{N}\sum_{n=1}^{N} c^{\mathrm{T}} Y_{\mathrm{n,T=1}}$	$\frac{1}{N}\sum_{n=1}^{N} c^{\mathrm{T}} Y_{\mathrm{n,T=1}} - \frac{1}{N}\sum_{n=1}^{N} Z_{\mathrm{n,T=1}}$
	$\frac{1}{N}\sum_{n=1}^{N} Z_{\mathrm{n,T=1}}$

Table 2: Balance Sheet at $T = 1$

The goal of this thesis is to maximize the sample counterpart of the expected surplus. This means, the goal is to maximize the following objective function:

$$\max_{c} \quad \frac{1}{N}\sum_{n=1}^{N} c^{\mathrm{T}} Y_{\mathrm{n,T=1}} - \frac{1}{N}\sum_{n=1}^{N} Z_{\mathrm{n,T=1}} \tag{1}$$

However, the second term in the maximization problem, $\frac{1}{N}\sum_{n=1}^{N} Z_{\mathrm{n,T=1}}$, does not depend on c^{T}. Therefore, Equation 1 can be rephrased in the following way:

$$\max_{c} \quad \frac{1}{N} c^{\mathrm{T}} \sum_{n=1}^{N} Y_{\mathrm{n,T=1}} \tag{2}$$

3.2 The Budget, the Shocks and the Constraints

Whenever an insurance company is investing into assets, it is subject to a budget constraint. Assuming that short sales as well as credit lines are not possible, the company can only invest as much money as it has in its bank account. In the mathematical notation of our model this means that at $T = 0$, the investments can be no more than the maximum amount of money available, B_0. Hence, we arrive at the first constraint of the model, the budget constraint, which can be written in the following way:

$$\mathrm{c^T Y_{T=0}} \leq B_0 \tag{3}$$

In order to test the robustness of the balance sheet of an insurer, Solvency II specifies a number of shocks that affect the surplus of the insurer. In this thesis, the shocks are limited to a change in the interest term structure, a change in the rate of mortality and a change in the value of equity. In order to make sure that the insurer remains in business, even if the surplus decreases, the surplus after each one of J shocks must be greater than 0. This suggests that a feasible investment strategy must satisfy the constraint:

$$c^{\mathrm{T}} Y_{\mathrm{n,j,T=1}} - Z_{\mathrm{n,j,T=1}} \geq 0 \ \forall \ economic\ states\ and \ \forall \ shocks \tag{4}$$

3.3 Re-phrasing the model as a Linear Program

Combining the objective function and the constraints from the subsections above yields the following linear program:

$$\begin{array}{lll} \text{maximize} & \frac{1}{N}\sum_{n=1}^{N} c^{\mathrm{T}} Y_{\mathrm{n,T=1}} & \\ \text{subject to} & & c^{\mathrm{T}} Y_{\mathrm{T=0}} \leq B0 \\ & & 0 \leq c_{\mathrm{i}} \ \forall \ c_{\mathrm{i}} \ in \ c \\ & & c^{\mathrm{T}} Y_{\mathrm{n,j,T=1}} - Z_{\mathrm{n,j,T=1}} \geq 0 \ \forall \ economic\ states\ and \ \forall \ shocks \end{array}$$

Based on this layout, the situation can be modeled using the programming language R.

4 The Toy Model - A Simplified Framework implemented in R

The entire programming of this thesis was completed in the program RStudio. For the implementation, the packages lpSolve, lpSolveAPI and ggplot2 were used. The packages lpSolve as well as lpSolveAPI use the Simplex algorithm to solve (constrained) linear programs.

4.1 Simplifying assumptions and Setup

This section presents a simplified framework of the model developed in the paper. The reason for including this conceptual model is not only to provide a basis for further expansion but also to illustrate the implementation of the theoretical layout from the previous section to the reader.
In order to make it easier to understand, the Toy Model makes a set of simplifying assumptions. Firstly, the investment universe has only 3 bonds with different maturities and yields. Secondly, it assumes that there are only 3 different economic scenarios. Each one of the economic scenarios generates a different interest term structure.

Also, there are only 2 shocks, namely a nominal interest increase along with a nominal interest rate decrease by 1%.
Figure 4 below shows the setup of the toy model in R. The model is solved with the help of the linear programming package "lpSolve-API", which is already implemented in R. Since the R-package that is used assumes the optimization problem to be a minimization problem, the objective function needed to be adjusted accordingly.

```
> ToyModel
Model name:
                          C1                  C2                  C3
Minimize    -626.15385143296   -1112.65138858803   -699.95607901696
R1          853.159069682688    1119.47657982157   857.342134238208  >=  938.297209261335
R2           285.14157994752    1125.83058722816   410.495177201664  >=  919.133370061965
R3           389.29601335296    1125.23825642701    507.90993852416  >=  922.540870832224
R4          1327.88456271667    1097.74907467981   1219.54730427392  >=  982.159707725868
R5          414.580686718976    1104.10308208845   565.104509907968  >=  961.568564054496
R6           579.80215804416    1103.51075128525    706.71960380416  >=  965.238272662909
R7          488.162791895552    1059.70196463411   611.186556150784  <=              1000
Kind                     Std                 Std                 Std
Type                    Real                Real                Real
Upper                    Inf                 Inf                 Inf
Lower                      0                   0                   0
```

Figure 4: Setup of Toy Model in R

Rows R1 to R6 correspond to the risk constraints and row R7 corresponds to the budget constraint. The final line correspond to the non-negativity constraint on the individual elements of c.

4.2 Optimal Allocation and Expected Surplus

The optimal allocation is shown in Figure 5 below.

```
> OptimalAllocation
[1] 0.0000000 0.7345919 0.235845
```

Figure 5: Optimal Allocation

As it can be seen, in Figure 5, the insurance company buys 0.73 units of the second bond and 0.24 units of the second bond at time $T = 0$.

4.3 Which constraints are binding?

In order to solve a linear program uniquely, the number of bonds that need to hold with equality is equal to the number of input variables. If a constraint holds with equality, it is said to be a binding constraint. The analysis shows that in the Toy Model, three constraints hold with equality. Hence, the model is uniquely identified. The binding constraints are the budget constraint, one non-negativity constraint as well as the first risk constraint from Figure 4. The binding risk constraint corresponds to an interest rate increase. The binding constraints can be considered as the economic scenarios/stresses that the portfolio is most vulnerable to. By this logic, the current model is most vulnerable to (1) an interest rate increase, (2) a change in the budget and (3) a change in the policy regarding short-selling.

4.4 How does the allocation perform under a new setup?

One relevant question to raise is how the optimal allocation changes, should the coefficients of the objective function and the risk constraints change. If the optimal allocation of a newly generated model is very dissimilar to the one computed before, the system is said to be unstable and minor changes in the economic situation could lead to a significant change in the optimal allocation, potentially resulting in the bankruptcy of the insurer. A stable solution, however, would be indicated by a negligible change in the optimal solution as well as by the old optimal solution being feasible under newly generated constraints.
One method to check the stability of the current setup is to re-build the model, recompute the coefficients of the objective function and solve the linear program. If the resulting allocation is significantly different to the one in the section before, the model is not stable. Figure 6 below shows the objective function as well as the constraints of the new model.

```
> ToyModel1
Model name:
                          C1                  C2                  C3
Minimize  -1237.92318578688     -725.377297152  -1798.48473101722
R1          1259.3136515584   563.602182521856   1783.1148135383  >=  862.375300021276
R2         1119.86334787379   673.105753872384   1851.59946784563  >=  797.847673838567
R3         1200.67436176589   608.130409279488   1810.81521304986  >=  835.434933945022
R4         1357.18528258048   782.379514891264   1749.73127928218  >=   900.92771517466
R5         1204.12354086093    944.58439721472    1818.2159335895  >=  832.148743910276
R6         1292.77082275021    848.26036411904   1777.43167879373  >=  872.193161206791
R7         1331.11057877811   620.650483705856    1587.5374197801  <=              1000
Kind                    Std                 Std                 Std
Type                   Real                Real                Real
Upper                   Inf                 Inf                 Inf
Lower                     0                   0                   0
```

Figure 6: Setup of second Toy Model in R

Figure 7 below shows both, the old optimal allocation as well as the newly generated optimal.

```
> OptimalAllocation
[1] 0.6136165 0.6609927 0.0000000
> OptimalAllocation1
[1] 0.000000 1.611213 0.000000
```

Figure 7: Comparison of the two optimal allocations

As it can be seen in Figure 7, the optimal allocation has changed significantly. The insurance company now buys 1.6 units of the second bond. Figure 8 shows whether or not the old optimal allocation satisfies the newly imposed constraints.

```
> ValidAllocation(OptimalAllocation,N,J,ConstMat1,ConstRHS1)
[1] "Optimal Allocation fails under new constraints."
[1] 0 0 0 0 0 0 1
```

Figure 8: Old optimal allocation does not satisfy the new constraints

Figure 8 does not only show that the old allocation violates the new constraints. It also specifies the exact constraint which is violated by the allocation. In this case, the old optimal allocation violates the budget constraint of the new setup. Based on the results

shown in Figures 7 and 8, one can conclude that the optimal allocation under this setup is in fact unstable.

5 Expanding the Simple Model

5.1 Addressing the instability of the optimal allocation

The main result of subsection 4.4 is that the optimal allocation of the Toy Model is unstable. One possible reason for the instability may be that the sample counterpart of the expected surplus is not yet close to its true value. The result of this would be that the coefficients of the variables in the objective function of the linear program are changed every time the economic situation change. A possible improvement would be to increase the number of economic scenarios. This would have two effects on the linear program: (1) The sample counterpart of expected surplus converges closer to its true value and (2) the number of Risk Constraints increases, which implies that the feasible region of the linear program becomes more refined. Hence, if the number of economic scenarios is increased from $N = 3$ to $N = 200$, the optimal solutions of two models should not only be similar, but also feasible under the constraints imposed by the other model. Figure 9 shows the optimal allocations for two models with a similar setup as the toy model.

```
> OptimalAllocationBiggerModel
[1] 0.0000000 0.8473681 0.2986264
> OptimalAllocationBiggerModel1
[1] 0.0000000 0.8606247 0.2934437
>
> ExpectedSurplusBiggerModel
[1] 1178.865
> ExpectedSurplusBiggerModel1
[1] 1227.573
```

Figure 9: Comparison of the two optimal allocations

The difference between allocations in Figures 7 and 9 is immediately obvious. The allocations in Figure 9 differ only by a marginal amount, whereas the differences between the allocations in Figure 7 are comparatively large. At the same time, tests indicate that the optimal allocations of the bigger models are feasible solutions under the constraints imposed by the other model. Based on the evidence, one can say that systems with more economic scenarios are truly more stable than systems with fewer economic scenarios. One can therefore deduce that increasing the number of economic states makes the sample counterpart of the expected surplus converge closer to its true value. This makes sense from a mathematical point of view. From a risk management perspective, however, the two scenarios are not very comparable. The "worst case scenario" among 10 economic scenarios may represent a one-in-ten-years recession. The "worst case scenario" among 1000 economic scenarios would by the same logic represent a one-in-thousand-years re-

cession. Which may be a lot worse.

5.2 Mortality shock

So far, the analysis has focused on a portfolio consisting of bonds, given constraints imposed by a budget, two interest rate shocks and a restriction on short-selling. However, as mentioned in section 2.2.1, it is important to not only take into account risks to the investment portfolio of the insurer, but also risks attached to the core business of an insurance company. Hence, this section exposes the setup resulting from section 5.1 to a 15% [3] increase in the mortality rate of the insured. The effect of this increase in the mortality is an increase in the number of claims to the insurer. This will not change the objective function of the model. However, it will impose new restrictions on the model via the risk constraints. An increase in the mortality rate by 15% makes the setup resulting from section 5.1 infeasible. However, increasing the budget of the insurance company at $T = 0$ from 1000 to 3000 results in a feasible solution to the linear program. The optimal allocation is shown in Figure 10 below.

```
> OptimalAllocation41
[1] 0.000000 2.987217 0.000000
```

Figure 10: Optimal Allocation

Based on the observations in this section, one can make several inferences. Firstly, the mortality risk appears to constrain the linear program more than the interest rate shocks. From an economic point of view, this makes sense, since the mortality of the customers is a risk, that is in itself not hedgable. Secondly, the section shows that increasing the amount of money on the asset side of the balance sheet is an effective measure to prevent an increase in the mortality from bankrupting the insurer.

5.3 Investment Universe, extension in economic states and new shock

In each economic state, the return of each index is randomly generated by a log-normal distribution with mean 2 and variance 0.16. To this point, this paper has analyzed an investment portfolio consisting only of bonds. In the real world, however, an insurance company can also invest in other assets, for instance, stocks. In order to make this analysis more realistic, this section will add two stock indices to the model. The model resulting from section 5.1 servers as a starting point for this section. At the end of that section, the insurance company could invest in 3 bonds at $T = 0$. 200 economic scenarios were used to derive both, an objective function with the sample counterpart of the expected value of each asset as a coefficient of the corresponding variable and 400 risk constraints. Adding two stock indices (Type 1 equity) will result in two additional variables in the objective function. Also, as explained in section 2.2.1 the equity shock (a decrease in the value of the stock index by 39% [3]) will result in 100 additional risk constraints. Including a new

asset class to the investment universe raises the question how the investment budget will be allocated between the two asset classes. Figures 11 and 12 show the proportion of the budget invested into bonds and stocks respectively.

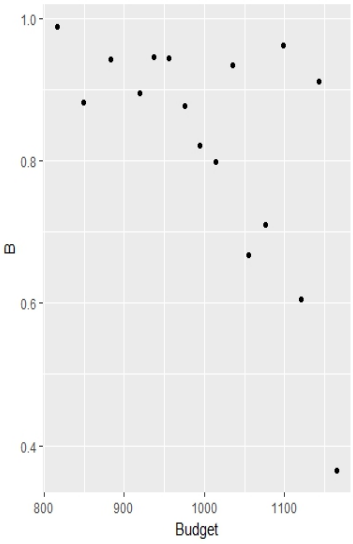

Figure 11: Bonds as a proportion of the investment budget vs Investment Budget

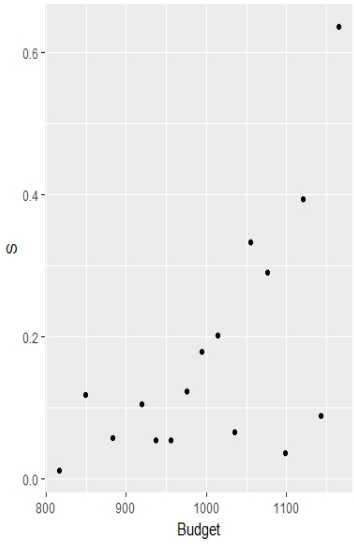

Figure 12: Stocks as a proportion of the investment budget vs Investment Budget

Figure 11 shows that for a budget of marginally more than 800 Euros, the portfolio of the insurance company consists almost entirely of bonds. Figure 11 also shows a clear downward trend in the proportion of bonds in the portfolio, as teh investment budget increases. This implies that as the investment budget of the insurer increases, the proportion of bonds in the portfolio decreases. A portfolio with a budget of about 1150 Euros, for instance, has less than 40% of its money invested in bonds.
Stocks on the other hand, make out a very low proportion of the insurers portfolio, whenever the investment budget is small. Figure 12 shows that for a budget of 800 Euros, the insurer has almost no stocks in his portfolio. Other than the proportion of bonds, the proportion of stocks in the portfolio has a clear upwards trend. As the company becomes allocates more money to its investment activities, the proportion of stocks in the portfolio increases. For a portfolio with an investment budget of 1150 Euros, the insurer invests more than 60% of his money in stocks.

6 Possible improvements

6.1 Is a linear programming approach a good choice for portfolio construction?

One question that can be raised is whether a linear program context is a good fit for optimizing an investment portfolio. Alexander Braun, Hato Schmeiser and Florian Schreiber

run a quadratic optimization program in their paper Portfolio Optimization Under Solvency II: Implicit Constraints Imposed by the Market Risk Standard Formula [1]. Also, the authors use classical portfolio theory in the construction of their portfolio. This may represent an opportunity for further extending the model.

6.2 Selection of interest rate term structure and stock returns

In the process of simulating different states of the economy, the program draws the interest rate term structure from a uniform distribution. The returns of the stock indices are drawn from a log-normal distribution. A more accurate approach to generating the interest rate term structure and the stock market returns may be using a bootstrapping approach. This would imply drawing the interest term structure and the returns from past observations, which could result in a more accurate representation of reality.

7 Conclusions

The analysis in this paper started off with a very simplified version of the model. This section showed that the constraints imposed by Solvency II do restrict the optimal allocation by imposing binding constraints on the allocation. The increase in the number of economic scenarios has lead to more stable optimal allocations. The shock in mortality has shown that the company can only survive some shocks by increasing the amount of money on the asset side of the balance sheet. Including stock indices has revealed that the proportion invested in one asset class can change with the size of the budget. The setup of the linear program can include all forms of constraints as well as other investments such as real estate, currency and much more. Hence, this analysis is not limited to insurers. The analysis presented in this paper can yield usable, relevant and easily understandable output for all kinds of portfolio managers. Thanks to the simplicity and the ease with which the program can be implemented, it can quickly be generalized.

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